

HAFIZ YUNUS KALATHIL

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EDUCATION

National Institute of Technology Karnataka
Bachelor of Technology, Mechanical Engineering, GPA: 8.23/10

Surathkal, India
2021 - 2025

SKILLS

Core Competencies Vibration Data Processing & Analysis, Robotics Mechanism Design, Bolted Joint Analysis, Structural FEA, Thermal Testing and Simulation, Automotive Mechanical Design
Tools Creo Parametric, SOLIDWORKS, Ansys (Mechanical, Fluent, Icepak), Simulink, Claude Code
Languages MATLAB, Arduino C, Python

PROFESSIONAL EXPERIENCE

Caterpillar Inc. **July 2025 - Present**
Associate Engineer - Genset Packaging Team *Bangalore, India*

- Part of the 2025–26 Engineering Rotational Development Program (ERDP), completing 3-month rotations at the Perkins Engine facility in Aurangabad and with the Genset NPI Program Management team in Bangalore.
- Moved to the Genset Package Analysis team, defining NPI validation test requirements and developing strain-gauge and vibration gauge maps for critical operating conditions and locations.
- Serving as a **vibration data analyst** across multiple NPI projects, comparing FEA predictions, existing-product test data, and new-product validation data to assess performance and define acceptance criteria.
- Completing test-to-report analysis within ~ 1 week under tight NPI timelines, enabling rapid validation of high-capacity gensets for growing data-center power demand.

Caterpillar Inc. **May 2024 - July 2024**
Mechanical Engineering Intern - Genset Packaging Team *Bangalore, India*

- Created a tool for genset base-rail design to optimize lift-eye placement, considering lift-eye forces and resulting stresses.
- Conducted bolted joint analysis of lift eyes and performed non-linear FEA of base rail assemblies using Creo Simulate.

Ergon Labs **June 2023 - July 2023**
Mechanical Engineering Intern *Bangalore, India*

- Conducted **thermal characterization** of MOSFET packages such as TO-247 and TO-220, analyzing results using a MATLAB script that reads a serial communication port and calculates thermal resistance and temperatures.
- Performed **parametric structural FEA** on heatsink clips for transistor packages using ANSYS Mechanical, considering the force applied and deformation required, generating around 100 designs.
- Conducted thermal analysis on the casing heatsink, optimizing fan placement and heatsink geometry to increase airflow and reduce temperatures using ANSYS Icepak and SOLIDWORKS Flow Simulation.

ACTIVITIES

NITKRacing (Formula Student Team) **June 2022 - March 2025**
Vehicle Dynamics Engineer & Mechanical Lead

- Led 25+ mechanical engineering students and **headed Formula Bharat 2026 mechanical design**, with hands-on responsibility across vehicle dynamics, suspension, brakes, vehicle modelling, gearbox design, and battery packaging.
- Achieved second place in Pi-EV 2024, the Formula Student concept challenge focused on EV powertrain design.
- Built the braking system for the vehicle, using braking force calculations and structural FEA.
- Developed a point mass lap time simulation solver on MATLAB for powertrain comparison and optimization.

CSD Robocon NITK (Competitive Robotics Team) **June 2022 - January 2024**
Mechanical Engineer & Head of Mechanical Subsystem

- Led a team of 10+ student mechanical engineers designing and manufacturing the robots for the competition.
- Qualified and participated in the national stage of ABU Robocon 2023 (An Asia-Pacific Robotics Competition).
- Designed and built a pair of **semi-autonomous robots**, capable of picking up, manipulating, and shooting soft rings.

PROJECTS

Design and Manufacturing of an Unmanned Surface Vessel (USV) **April 2024 - April 2025**

- Developed the mechanical systems for an autonomous canoe for search and rescue, and surveying applications.
- Built MATLAB simulations of canoe dynamics and PID control systems for autonomous waypoint navigation.
- Designed and manufactured a trailer supporting one-person operation for loading and deploying a 500 kg autonomous canoe, featuring an electric winch and roller ramp.

Design and Manufacturing of a Combat Robot **September 2023 - October 2024**

- Designed and manufactured a **15kg combat robot** with a vertical bar spinner, modular aluminium chassis, and a unique chain-driven four-wheeled drive system.
- Competed in several national-level competitions, achieving second place at TechTatva 2023, MIT Manipal.